

Practice Problem Set 2

ECON 308 – International Economic Relations

Fall 2025

Short Conceptual Questions

1. Explain how an increase in a country's labor endowment affects its production possibilities frontier and pattern of trade under the Heckscher–Ohlin model.
2. Define the **Rybczynski theorem**. What does it predict about output levels when one factor endowment increases, holding goods prices constant?
3. Suppose capital is relatively scarce in Country A compared to Country B. According to the Stolper–Samuelson theorem, what happens to the real wage and real rental rate in Country A when it opens to trade?
4. Distinguish between **export-biased growth** and **import-biased growth**. How does each affect a country's terms of trade?
5. In the Standard Trade Model, what is meant by an “improvement in the terms of trade”? How does it influence national welfare?
6. Define **external economies of scale**. How do they differ from **internal economies of scale** in terms of market structure and trade patterns?
7. Briefly describe how **path dependence** can lead to persistent differences in industrial location under external economies.

Analytical and Numerical Problems

Problem 1: Long-Run Factor Adjustment (Heckscher–Ohlin Model)

Two countries, Home (H) and Foreign (F), produce two goods: textiles (T) and steel (S). Textiles are **labour-intensive**, and steel is **capital-intensive**.

$$K_H = 400, \quad L_H = 800, \quad K_T/L_T = 0.25, \quad K_S/L_S = 2.$$

- (a) Compute Home's relative factor abundance.
- (b) Which good will Home export under free trade?
- (c) Using the Stolper–Samuelson theorem, predict how trade affects real wages and real returns to capital in Home.
- (d) Suppose the world price of textiles rises by 10%. Explain qualitatively how factor prices will adjust.

Problem 2: The Standard Trade Model and Welfare

A country produces food (F) and electronics (E) with a concave PPF given by:

$$F^2 + 2E^2 = 200.$$

The world relative price is $P_E/P_F = 2$.

- (a) Determine the production point that maximizes the value of output.
- (b) If consumption preferences imply $F = E$, find the consumption bundle and trade pattern.
- (c) Evaluate qualitatively whether the country gains from trade compared to autarky using indifference curves and the terms of trade concept.
- (d) Suppose the terms of trade improve (export price rises). Explain the welfare effect using a diagrammatic argument.

Problem 3: External Economies and Industrial Location

Consider two identical countries, A and B, producing semiconductors under **external economies of scale**. Industry output in a country determines unit cost:

$$C = 20 - 0.05Q,$$

where Q is total industry output in that country. World demand is:

$$Q^W = 400 - 5P.$$

- (a) Derive the relationship between price P and total world output Q^W .
- (b) Suppose initially each country produces half of world output. Compute the equilibrium price.
- (c) Show that if one country produces slightly more, its average cost falls—illustrate why this can lead to complete specialization in one location.
- (d) Explain how such specialization can result in trade even between identical countries.
- (e) Discuss how government policy (e.g., temporary subsidy or protection) could shift production location permanently.